

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \\ -4 \\ 3 \end{pmatrix}.$$

$$\hat{u} = \frac{1}{\sqrt{30}} \begin{pmatrix} -1 \\ 2 \\ -4 \\ 3 \end{pmatrix}.$$

a =

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{pmatrix}.$$

$$a \bullet b = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{pmatrix} \bullet \begin{pmatrix} 9 \\ 7 \\ 0 \\ 1 \\ 0 \\ 0 \\ 8 \\ 9 \\ 0 \end{pmatrix} = 155.$$

$$\mathbf{u} = \begin{pmatrix} -2 \\ 3 \\ 1 \\ 4 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix}.$$

$$P_0 - Q = \begin{pmatrix} x_0 - x_1 \\ y_0 - y_1 \\ z_0 - z_1 \end{pmatrix}, \quad (P_0 - Q) \bullet \mathfrak{n} = a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1),$$

$$P_0 = \begin{pmatrix} 1 \\ -4 \\ -3 \end{pmatrix}$$

$$m_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \quad y \quad m_2 = \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix},$$

$$W = \left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \in \mathbb{R}^4 \mid 2x - y + 3z - w = 0 \right\}.$$

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = x \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$$W =_{\text{gen}} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

$$\hat{u}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}.$$

$$\mathbb{w}_2 = \mathbb{v}_2 - (\mathbb{v}_2 \bullet \hat{\mathbb{w}}_1) \hat{\mathbb{w}}_1$$

$$= \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} - \left[\begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} \bullet \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right] \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{5} \left[\begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right] \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \end{pmatrix} - \frac{6}{5} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}$$

$$\hat{\mathbb{W}}_2 = \frac{1}{\sqrt{70}} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}.$$

$$\mathbb{W}_3 = \mathbb{V}_3 - (\mathbb{V}_3 \bullet \hat{\mathbb{U}}_1) \hat{\mathbb{U}}_1 - (\mathbb{V}_3 \bullet \hat{\mathbb{U}}_2) \hat{\mathbb{U}}_2$$

$$= \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} - \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \bullet \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right] \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \\ - \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \bullet \frac{1}{\sqrt{70}} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix} \right] \frac{1}{\sqrt{70}} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{5} \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \bullet \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \right] \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{70} \left[\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \bullet \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix} \right] \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} + \frac{2}{5} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + \frac{3}{70} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}$$

$$= \frac{1}{14} \begin{pmatrix} 2 \\ -1 \\ 3 \\ 14 \end{pmatrix}$$

$$\hat{w}_3 = \frac{1}{\sqrt{210}} \begin{pmatrix} 2 \\ -1 \\ 3 \\ 14 \end{pmatrix}.$$

$$\mathcal{B} = \left\{ \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{70}} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{210}} \begin{pmatrix} 2 \\ -1 \\ 3 \\ 14 \end{pmatrix} \right\}.$$

$$v_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

$$\mathbb{w}_4 = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 3 \\ -1 \end{pmatrix},$$

$$\hat{\mathbb{w}}_4 = \frac{1}{\sqrt{15}} \begin{pmatrix} 2 \\ -1 \\ 3 \\ -1 \end{pmatrix}.$$

$$\mathcal{B}'' = \left\{ \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{70}} \begin{pmatrix} -6 \\ 3 \\ 5 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{210}} \begin{pmatrix} 2 \\ -1 \\ 3 \\ 14 \end{pmatrix}, \frac{1}{\sqrt{15}} \begin{pmatrix} 2 \\ -1 \\ 3 \\ -1 \end{pmatrix} \right\}.$$

$$\mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix},$$

$$\text{proy}_W \mathbf{v} = (\mathbf{v} \bullet \hat{\mathbf{u}}_1) \hat{\mathbf{u}}_1 + (\mathbf{v} \bullet \hat{\mathbf{u}}_2) \hat{\mathbf{u}}_2 + (\mathbf{v} \bullet \hat{\mathbf{u}}_3) \hat{\mathbf{u}}_3 = \frac{1}{5} \begin{pmatrix} -1 \\ 3 \\ 1 \\ -2 \end{pmatrix}.$$